Moving to Edmonton Made Easy

Aaron A, Omar M, Ryley G, Vlad M

The Problem and Overview

Problem

It is often difficult for immigrants or people within the city to find a suitable place to live.

Motivation

To help immigrants or people within the city to find suitable neighbourhoods to live in without the hassle of going through multiple resources.

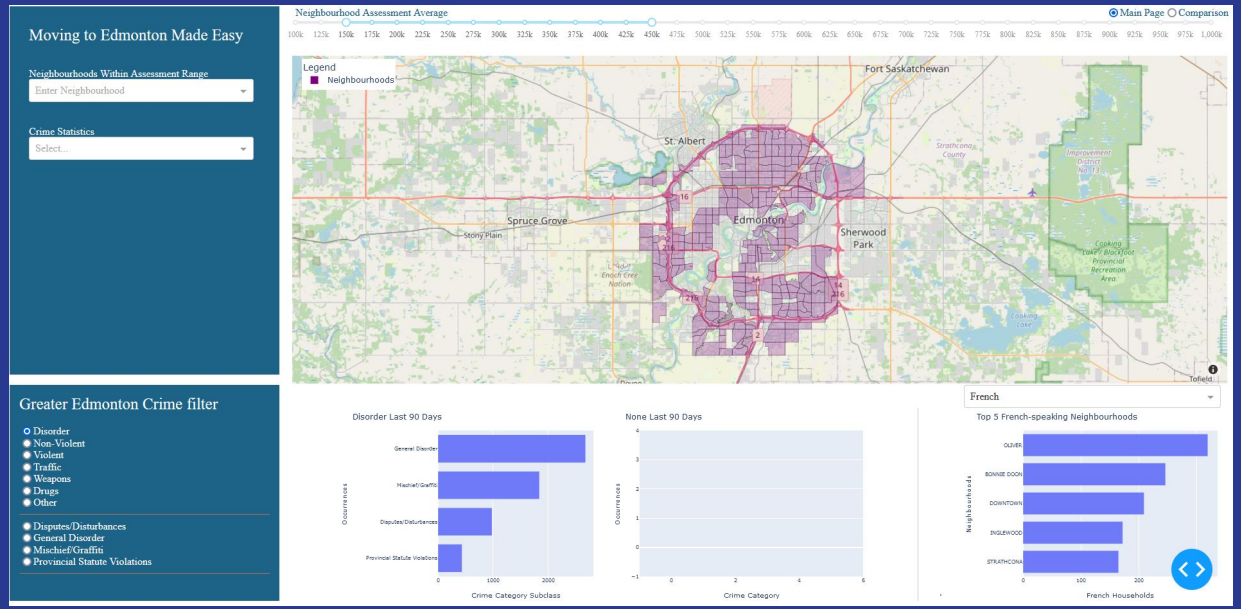
Benefits

- User can find suitable neighbourhoods to live in more faster.
- Users can determine whether a neighbourhood is safe to live in according to their standards.
- Immigrants can find people with the same background.

Implementation

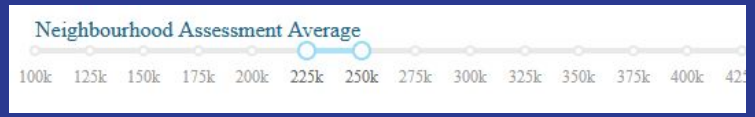
We decided to make an interactive dashboard using Plotly Dash to visualize data useful for our target users.

Dashboard overview

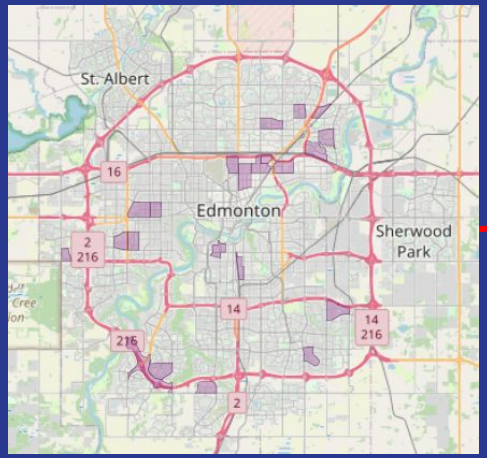


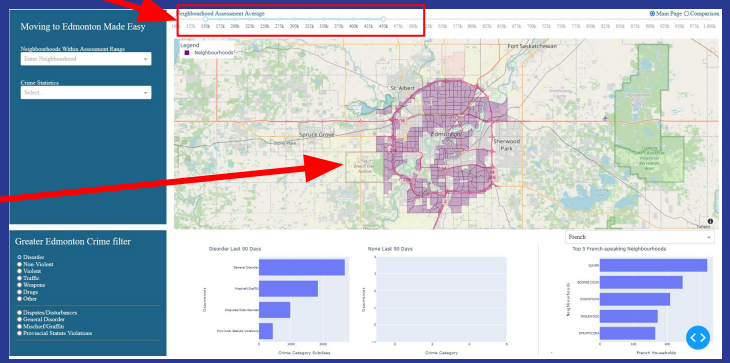
Filtered Map component with information bar charts

Average Assessment per Neighbourhood

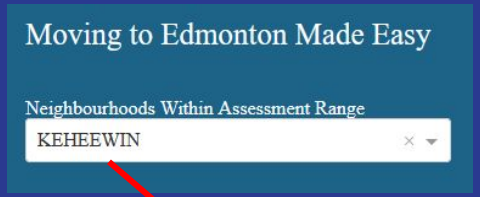


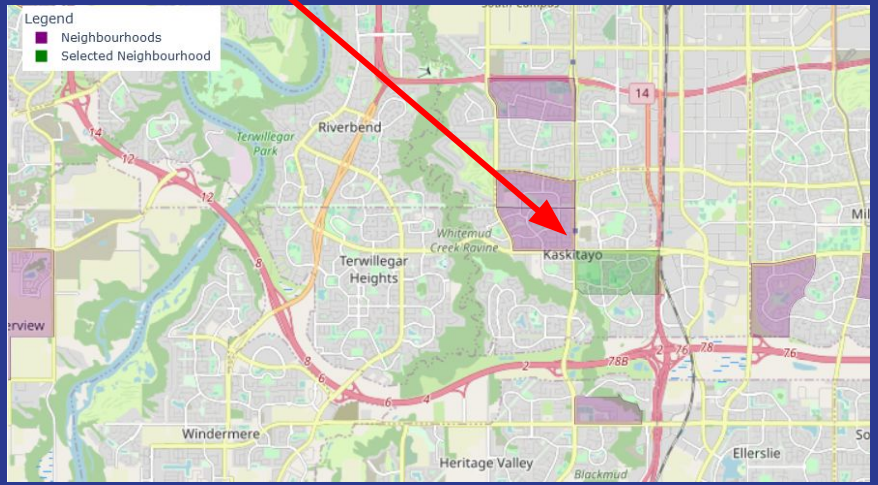
- Slider will filter and highlight only the neighbourhoods within that range





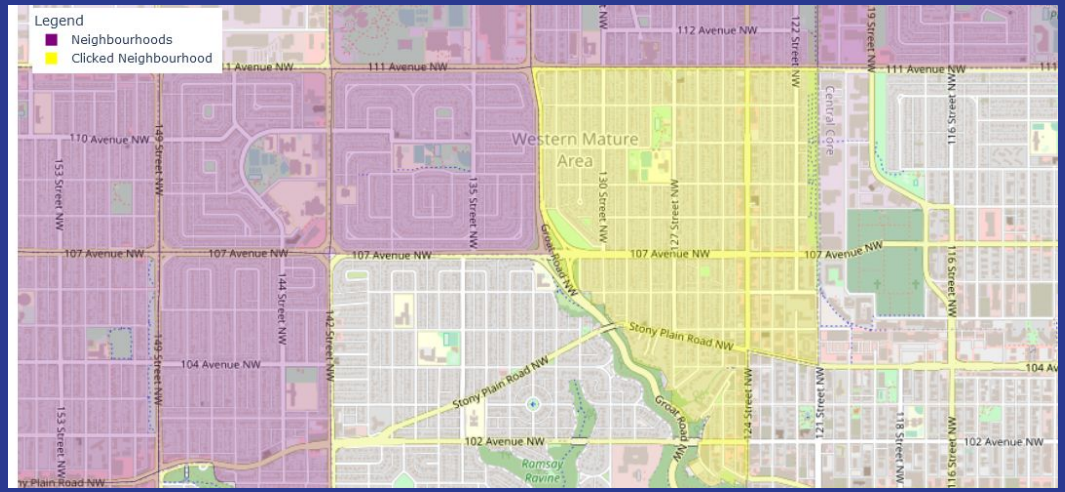
Neighbourhood Selection





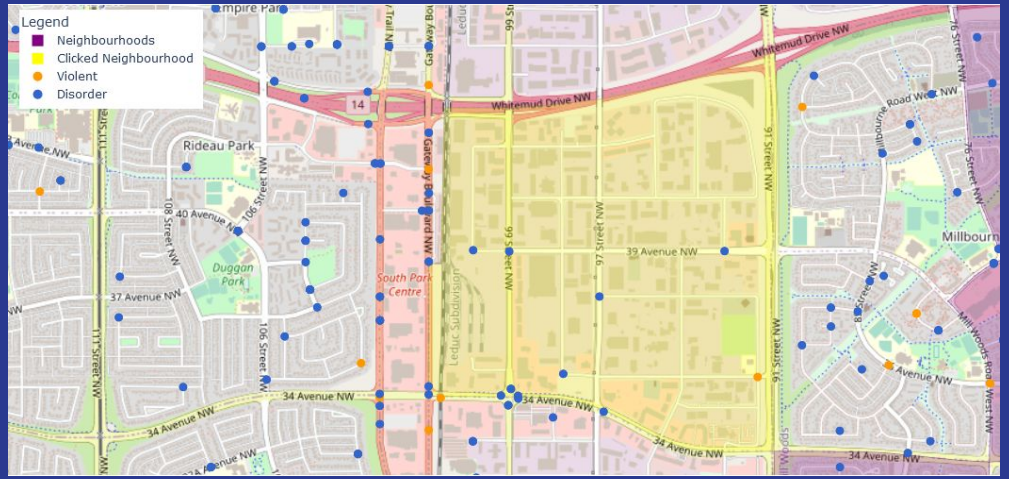
- The selection dropdown will contain available neighbourhoods within that range
- Once selected, the map will highlight the neighbourhood in green

Neighbourhood Selection



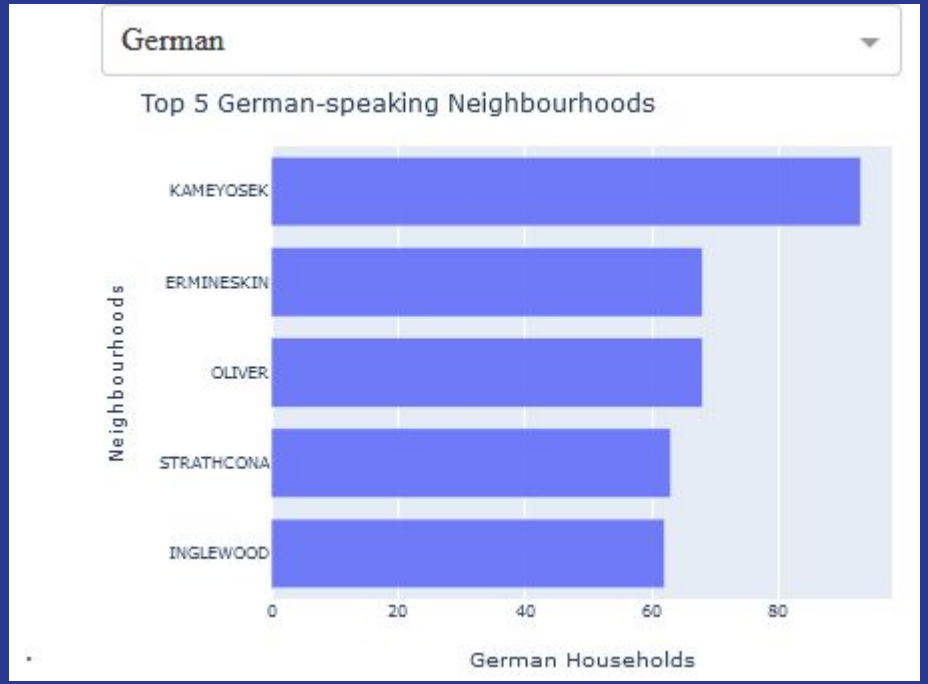
- Clicking on a neighbourhood will highlight the neighbourhood in yellow, and zoom in closer to it.
- Double clicking the map will reset it to the whole city view.

Display Crime Occurrences



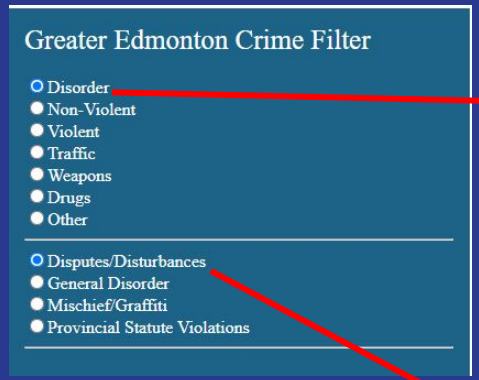
- The multiple dropdown shows the user where occurrences took place by category.
- This will allow the user to view high risk areas.

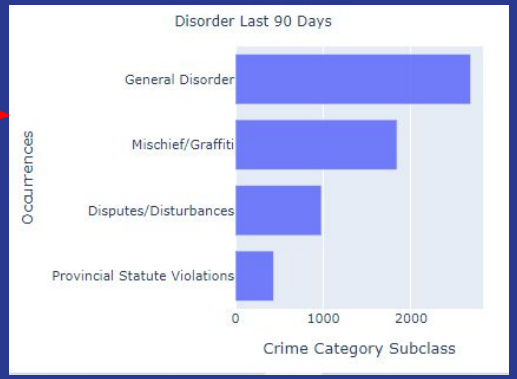
Language Chart



- Languages dropdown will display to the user the top five neighbourhoods where the chosen language is spoken in number of households.
- This provides insight to the user about the community cultural demographics, as well as potential cultural resources for the user.

Greater Edmonton Crime Filter

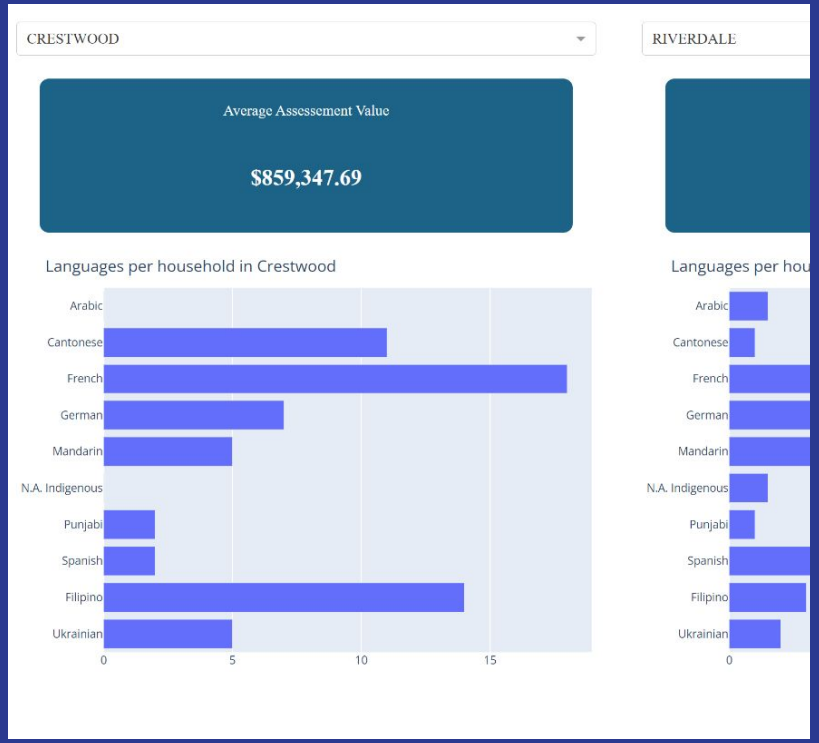




- The user has the ability to select different crime types rapidly.
- General crime categories are on top and the crime subcategories are on the bottom.

- Users are not only interested in what crimes are in neighbourhoods but want to know overall crime, because most will be traveling throughout the city.

Comparison Dashboard



- The user can select 3 neighbourhoods from the dropdown list and compare between them.
- Each selection will show the average assessment value for the neighbourhood, and the number of households speaking a different language.

- Users can compare between 3 neighbourhoods the average value for a property, and the languages spoken for current residents.

Preparing and Processing Data

Language Dataset

Major Non-English Languages

- Disregarded English
- Disregarded “Other” languages
- Renamed column names

Property Assessment Dataset

Average Property Assessments

- Calculated average residential property prices
- Filtered assessment values
- Renamed column names

90-day Crime Occurrences Dataset

Categorized Crimes

- Categorized crimes
- Calculated crime numbers

Insightful Information

1. The neighbourhoods *Oliver*, *Boone Doon*, and *Downtown* have the highest amount of households speaking French.
2. [Property] Theft under $5000 is the most reported crime with over 6K reports.
3. The neighbourhood Ermineskin has an average assessment value of $138K, with Filipino being spoken in 160 households.

Limitations

Challenge 1

Relating the crime data to a specific neighbourhood

- The crime occurrences have locations only, not Neighbourhood data
- Google API does not return neighbourhood.

Challenge 2

Map used does not cluster crime data

- Displays the entire crime category

Challenge 3

Languages info on hover

- Neighbourhood doesn’t show the most spoken language on hover

DEMO